SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – Coding Challenge

Course Code:	**DSA0405**	**Course Title:**	**FUNDAMENTALS OF DATA SCIENCE**
CO Assessed:	**CO5 – Coding Challenge**	**Assessment:**	**Assessment Tool 1– Coding Challenge**
Weightage:	**40% of CIA**	**Total Marks:**	**100**
CO5 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	Abzam A	**Reg. No.:**	192573029
Section / Batch:	CSE - DS	**Date:**	27/08/26

Code of Conduct

I Abzam A (192573029) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Abzam A

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	**25**	**Thorough understanding; clearly identifies all key issues**	**Good understanding with most issues identified**	**Basic understanding; some issues missed**	**Poor understanding; problem not identified correctly**	
Application of Concepts	**25**	**Effectively applies relevant concepts to analyze the case deeply**	**Applies appropriate concepts with minor gaps**	**Limited application of concepts**	**Fails to apply relevant concepts**	
Analysis	**20**	**In-depth analysis with strong reasoning and insights**	**Good analysis with logical reasoning**	**Basic analysis with limited depth**	**Weak or no analysis; lacks reasoning**	

Solution Design	**20**	**Innovative, feasible solutions with strong justification**	**Logical solutions with adequate justification**	**Basic solutions with weak justification**	**Incorrect or no solution provided**	
Clarity	**10**	**Clear, well-structured, and easy to understand**	**Organized and understandable presentation**	**Limited clarity and structure**	**Poor clarity; difficult to understand**	

DSA0404 – Fundamentals of Data Science
CO5-AT2 Coding Challenge

Q1. Design and implement a custom evaluation pipeline that compares kNN, Decision Tree, Logistic Regression, and K-Means-derived cluster labels (used as a weak classifier) on the same dataset using a statistically valid comparison —implement a paired t-test or McNemar's test from scratch across cross-validation folds to determine whether the performancedifferences between models are statistically significant, not just numerically different.

Code:

```
import numpy as np
from math import sqrt, erf
from sklearn.datasets import load_breast_cancer
from sklearn.model_selection import StratifiedKFold
from sklearn.preprocessing import StandardScaler
from sklearn.neighbors import KNeighborsClassifier
from sklearn.tree import DecisionTreeClassifier
from sklearn.linear_model import LogisticRegression
from sklearn.cluster import KMeans
from sklearn.metrics import accuracy_score



def normal_cdf(x):
    return 0.5 * (1 + erf(x / sqrt(2)))



def paired_t_test(scores1, scores2):
    differences = np.array(scores1) - np.array(scores2)
```

```
    n = len(differences)
    mean_difference = np.mean(differences)
    std_difference = np.std(differences, ddof=1)

    if std_difference == 0:
        t_statistic = 0
        p_value = 1.0
    else:
        t_statistic = mean_difference / (std_difference / sqrt(n))
        p_value = 2 * (1 - normal_cdf(abs(t_statistic)))

    return t_statistic, p_value



data = load_breast_cancer()
X = data.data
y = data.target

cv = StratifiedKFold(n_splits=10, shuffle=True, random_state=42)

knn_scores = []
dt_scores = []
lr_scores = []
kmeans_scores = []

for fold, (train_index, test_index) in enumerate(cv.split(X, y), start=1):
```

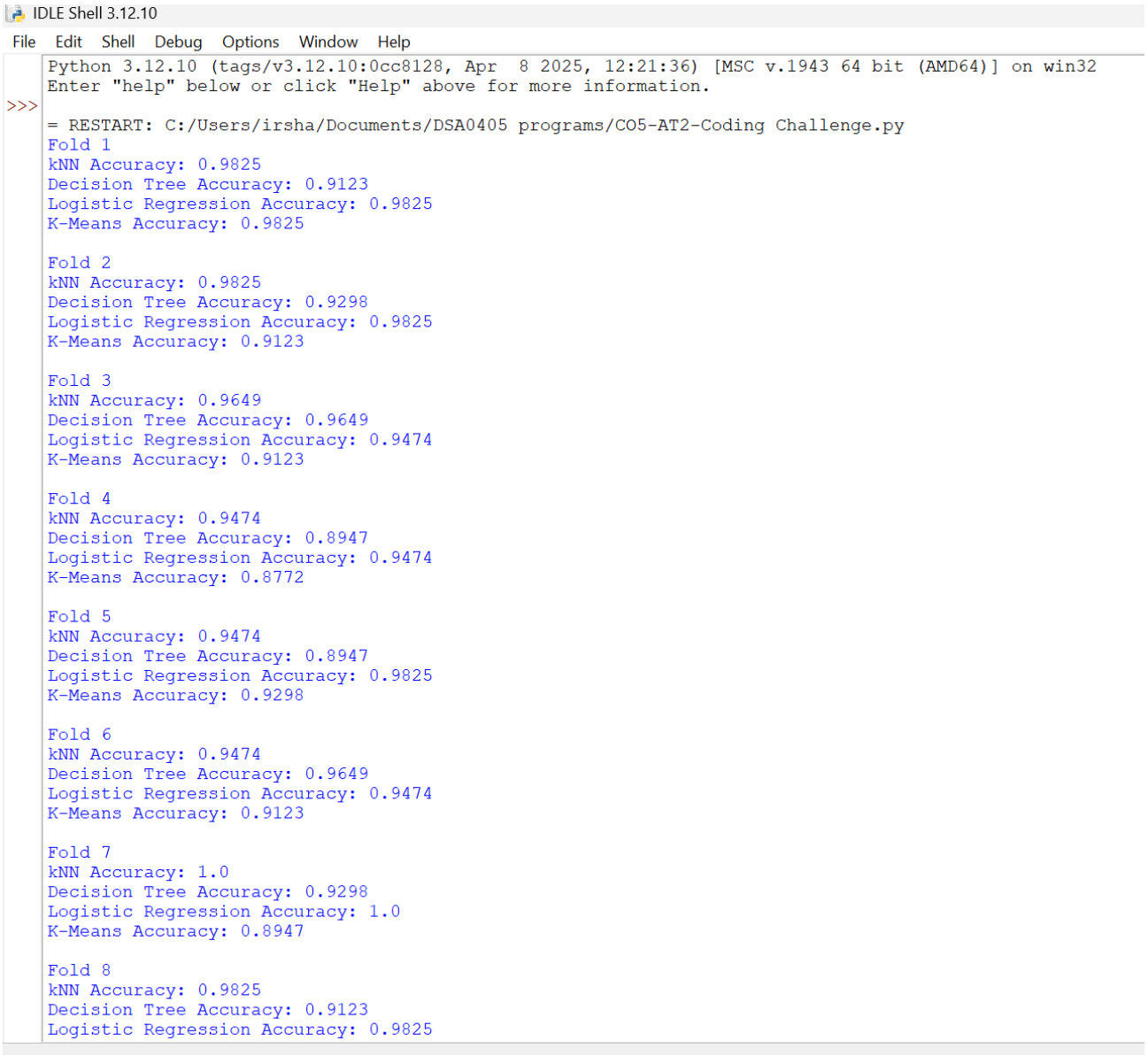
```
X_train = X[train_index]
X_test = X[test_index]
y_train = y[train_index]
y_test = y[test_index]

scaler = StandardScaler()
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)

knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train_scaled, y_train)
knn_predictions = knn.predict(X_test_scaled)
knn_accuracy = accuracy_score(y_test, knn_predictions)
knn_scores.append(knn_accuracy)

decision_tree = DecisionTreeClassifier(random_state=42)
decision_tree.fit(X_train, y_train)
dt_predictions = decision_tree.predict(X_test)
dt_accuracy = accuracy_score(y_test, dt_predictions)
dt_scores.append(dt_accuracy)

logistic_regression = LogisticRegression(max_iter=5000, random_state=42)
logistic_regression.fit(X_train_scaled, y_train)
lr_predictions = logistic_regression.predict(X_test_scaled)
lr_accuracy = accuracy_score(y_test, lr_predictions)
lr_scores.append(lr_accuracy)
```

```
kmeans = KMeans(n_clusters=2, random_state=42, n_init=10)
kmeans.fit(X_train_scaled)

train_clusters = kmeans.predict(X_train_scaled)
cluster_to_class = {}

for cluster in range(2):
    cluster_labels = y_train[train_clusters == cluster]

    if len(cluster_labels) > 0:
        cluster_to_class[cluster] = np.bincount(cluster_labels).argmax()
    else:
        cluster_to_class[cluster] = 0

test_clusters = kmeans.predict(X_test_scaled)

kmeans_predictions = np.array(
    [cluster_to_class[cluster] for cluster in test_clusters]
)

kmeans_accuracy = accuracy_score(y_test, kmeans_predictions)
kmeans_scores.append(kmeans_accuracy)

print(f"Fold {fold}")
print("kNN Accuracy:", round(knn_accuracy, 4))
print("Decision Tree Accuracy:", round(dt_accuracy, 4))
print("Logistic Regression Accuracy:", round(lr_accuracy, 4))
```

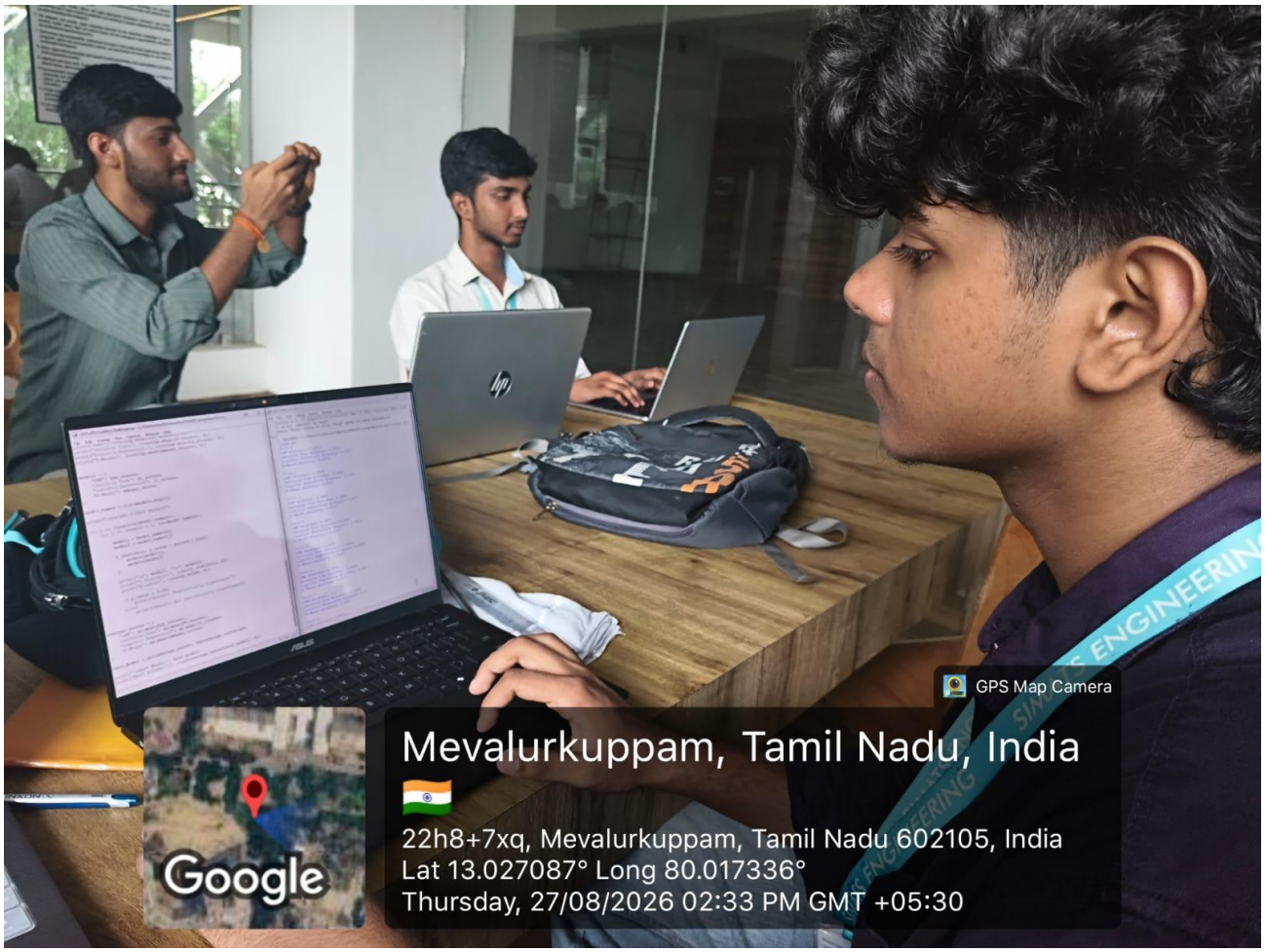
```
    print("K-Means Accuracy:", round(kmeans_accuracy, 4))
    print()


print("AVERAGE ACCURACY")
print("kNN:", round(np.mean(knn_scores), 4))
print("Decision Tree:", round(np.mean(dt_scores), 4))
print("Logistic Regression:", round(np.mean(lr_scores), 4))
print("K-Means:", round(np.mean(kmeans_scores), 4))


models = {
    "kNN": knn_scores,
    "Decision Tree": dt_scores,
    "Logistic Regression": lr_scores,
    "K-Means": kmeans_scores
}

model_names = list(models.keys())

print("\nPAIRED T-TEST RESULTS")

for i in range(len(model_names)):
    for j in range(i + 1, len(model_names)):

        model1 = model_names[i]
        model2 = model_names[j]
```

```
        t_statistic, p_value = paired_t_test(
            models[model1],
            models[model2]
        )

        print("\n", model1, "vs", model2)
        print("t-statistic:", round(t_statistic, 4))
        print("p-value:", round(p_value, 4))

        if p_value < 0.05:
            print("Result: Statistically Significant")
        else:
            print("Result: Not Statistically Significant")



average_scores = {
    "kNN": np.mean(knn_scores),
    "Decision Tree": np.mean(dt_scores),
    "Logistic Regression": np.mean(lr_scores),
    "K-Means": np.mean(kmeans_scores)
}

best_model = max(average_scores, key=average_scores.get)

print("\nBEST MODEL:", best_model)
print("BEST ACCURACY:", round(average_scores[best_model], 4))
```

Output:


IDLE Shell 3.12.10
File Edit Shell Debug Options Window Help
Python 3.12.10 (tags/v3.12.10:0cc8128, Apr 8 2025, 12:21:36) [MSC v.1943 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>>
= RESTART: C:/Users/irsha/Documents/DSA0405 programs/CO5-AT2-Coding Challenge.py
Fold 1
kNN Accuracy: 0.9825
Decision Tree Accuracy: 0.9123
Logistic Regression Accuracy: 0.9825
K-Means Accuracy: 0.9825
Fold 2
kNN Accuracy: 0.9825
Decision Tree Accuracy: 0.9298
Logistic Regression Accuracy: 0.9825
K-Means Accuracy: 0.9123
Fold 3
kNN Accuracy: 0.9649
Decision Tree Accuracy: 0.9649
Logistic Regression Accuracy: 0.9474
K-Means Accuracy: 0.9123
Fold 4
kNN Accuracy: 0.9474
Decision Tree Accuracy: 0.8947
Logistic Regression Accuracy: 0.9474
K-Means Accuracy: 0.8772
Fold 5
kNN Accuracy: 0.9474
Decision Tree Accuracy: 0.8947
Logistic Regression Accuracy: 0.9825
K-Means Accuracy: 0.9298
Fold 6
kNN Accuracy: 0.9474
Decision Tree Accuracy: 0.9649
Logistic Regression Accuracy: 0.9474
K-Means Accuracy: 0.9123
Fold 7
kNN Accuracy: 1.0
Decision Tree Accuracy: 0.9298
Logistic Regression Accuracy: 1.0
K-Means Accuracy: 0.8947
Fold 8
kNN Accuracy: 0.9825
Decision Tree Accuracy: 0.9123
Logistic Regression Accuracy: 0.9825

Geotag:


GPS Map Camera
Mevalurkuppam, Tamil Nadu, India
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Lat 13.027087° Long 80.017336°
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